

DATA CARD

Project: Optimising RAG Document Retrieval for Agronomic Advice

Team Albert

Input vs. Output

What are users inputting? What are the outputs or labels your model will predict or generate?

Users input a text-based agricultural question, for example, a farmer asking why their crop's leaves are turning yellow, or how to manage flooding on their farm. The system does not generate a written answer. Instead, it outputs a ranked list of the 5 most relevant document IDs from our knowledge base of 695 agricultural factsheets, ordered from most to least relevant. The retrieval system matches queries against each document's title and text fields to produce this ranking.

Classification Problem

What is the classification task, if any? What categories or classes are you working with, and why?

This is not a classification task, it is an information retrieval and ranking task. Rather than predicting a category, the system ranks all 695 documents in the corpus against a given query and selects the top 5. Retrieval quality is evaluated using graded relevance judgements on a 0 to 3 scale (0 = not relevant, 1 = low relevance, 2 = relevant, 3 = highly relevant), and performance is measured using nDCG at 5, which rewards placing the most relevant documents near the top of the ranking, not just including them somewhere in the top 5.

Labelling

Where do your labels come from? How will you check for consistency and accuracy?

Our relevance labels come from the `qrels_train.csv` file provided through the Kaggle competition, containing 4,194 query-document relevance judgements. We do not have information confirming whether these were created by agricultural domain experts, crowd workers, or another method, since this is not disclosed in the competition data available to us, so we treat them as the official evaluation labels without overstating their provenance. To check for consistency, we apply the same relevance definitions throughout our own local evaluation, and we plan to inspect queries where our retrieval performs poorly to identify any patterns suggesting label inconsistency.

Data Creation

How will you collect your data? Will it come from direct sources, or will you synthesize data in low-resource settings?

We are not collecting new data ourselves. We are using the fixed dataset supplied through the Kaggle competition, consisting of 695 agricultural documents, 308 training queries, 4,194 relevance judgements, and 200 test queries. On inspecting the data directly, we found that document origin is recorded as either "synthetic" (637 documents) or "llm_grounded" (58 documents), meaning the majority of the corpus was

synthetically generated rather than sourced from direct interviews or recordings, with a smaller portion grounded in real reference material.

Web Collection Ethics

Do you plan to source data from the web? Have you addressed consent, copyright, and privacy?

We are not sourcing any additional data from the web for this project, and are working entirely within the dataset supplied through the competition. On inspecting the license field directly, we found that licensing is split between "synthetic (CC0)" (637 documents) and a small number of real open licenses, including CC-BY-4.0 (55), CC-BY-SA-4.0 (1), CC0-1.0 (1), and CC-BY-3.0 (1). Since we are not scraping the web or collecting data from individuals ourselves, there is no new consent, copyright, or privacy exposure introduced by our project beyond what the competition dataset already carries.

Bias and Representation

Who might be left out of your dataset? How will you ensure it reflects the diversity of the community?

Our dataset covers 21 countries across Sub-Saharan Africa, but representation is uneven. Nigeria is the most represented single country with 133 documents (about 19 percent of the corpus), followed by Ghana (103), Tanzania (92), and Kenya (75), while countries like Togo (3) and Benin (4) are represented by only a handful of documents. This means our retrieval system is likely to perform more reliably for better represented countries and crops than for underrepresented ones. There may also be a gap between how farmers naturally phrase a problem and how it is worded in these factsheets, which is part of why we are testing both keyword-based and meaning-based retrieval methods, to reduce reliance on exact wording matches. We plan to review queries where retrieval performs poorly to check whether underrepresented countries or crops are disproportionately affected.

Values and Community Alignment

How does your dataset design reflect your values? How will your community benefit, and how will you ensure accountability?

This dataset design supports the values we identified earlier: accuracy over convenience, accessibility of existing knowledge, caution around unintended harm, and groundedness in real, verifiable sources. Because a majority of the documents are synthetically generated, our commitment to groundedness means being transparent about this rather than presenting the corpus as entirely authoritative published material. Our community, smallholder farmers and the extension workers who support them, benefits from a system that helps existing agricultural knowledge become easier to find, without treating our retrieval output as a substitute for expert judgement. We maintain accountability by documenting exactly where our data comes from, how relevance is labelled and evaluated, and where our dataset's coverage is strongest or weakest, so that any future use of this system is grounded in a clear understanding of its actual limitations.